

ORIGINAL PAPER

A discrete event simulation to evaluate impact of radiology process changes on emergency department computed tomography access

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Abstract

Background: Hospitals face the challenge of managing demand for limited computed tomography (CT) resources from multiple patient types while ensuring timely access.

Methods: A discrete event simulation model was created to evaluate CT access time for emergency department (ED) patients at a large academic medical center with six unique CT machines that serve unscheduled emergency, semi-scheduled inpatient, and scheduled outpatient demand. Three operational interventions were tested: adding additional patient transporters, using an alternative creatinine lab, and adding a registered nurse dedicated to monitoring CT patients in the ED.

Results: All interventions improved access times. Adding one or two transporters improved ED access times by up to 9.8 minutes (Mann-Whitney (MW) CI: [−11.0, −8.7]) and 10.3 minutes (MW CI [−11.5, −9.2]). The alternative creatinine and RN interventions provided 3-minute (MW CI: [−4.0, −2.0]) and 8.5-minute (MW CI: [−9.7, −8.3]) improvements.

Conclusions: Adding one transporter provided the greatest combination of reduced delay and ability to implement. The projected simulation improvements have been realized in practice.

KEYWORDS

access time, discrete event simulation, patient flow, process improvement, radiology

1 | INTRODUCTION

Computed Tomography (CT) is a vital diagnostic tool in clinical settings to visualize nearly all parts of the body and diagnose disease or injury, as well as to plan medical, surgical or radiation treatment.^{1,2} CT use in the emergency department (ED) has substantially increased in the past couple of decades^{2,3} and delays in imaging can result in serious negative health outcomes.^{4,5} In acknowledgement of this, national guidelines have been set for particularly emergent patients, such as stroke patients, who should receive a CT within 25 minutes of arriving at the ED.⁶ To provide timely care, a hospital must find ways to maximize the usage of scanners and minimize the time patients wait for

those scanners, while dealing with the challenge of managing semi-scheduled inpatients, scheduled outpatients, and unscheduled ED patients, all of whom have need for the same pool of medical imaging resources. In addition, the operational cost per minute a patient is in the emergency department has been estimated to be \$2.50, translating to \$250 000 in annual savings for a healthcare system that simply reduces their ED to imaging time for 10 000 CT patients by ten minutes.⁷

In such complex operations, interventions ranging from changing the process flow to adding staff can improve processes. However, many of these interventions are costly to implement, especially if the magnitude of impact the intervention will have on process



improvement is unknown. One way to address this is to use computer simulation to evaluate the effect on key outcome measures before implementing the interventions in practice. This can be accomplished preemptively, with no interruption to the practice, without large upfront capital expenditures, and can be applied in radiology practices to assess changes in patient arrival rates, available personnel, number of equipment, scheduling policies, etc.^{5,8-15}

This study aims to simulate potential process changes in the Department of Radiology at Saint Marys Hospital, which is a campus of the Mayo Clinic—a quaternary academic medical center located in Rochester, Minnesota. The Department of Radiology funded the Radiology Clinical Engineering Learning Laboratory (RadCELL) to bring together engineers and practitioners in the pursuit of operational improvements within the department. This study focuses on RadCELL's research into using discrete event simulation (DES) to model and improve ED patient access time within the department's CT modality, which houses six unique CT machines that serve emergency, inpatient, and outpatient demand.

DES is a method of replicating the operation of a real-world system through computer models that allow the user to gain evidence-based insights on the impact of changes to the system prior to implementation. These models focus on entities (eg, patients) that arrive to, move throughout, and exit the system according to logical rules of the clinical practice. While in the system, entities may require the use of one or more resources (eg, hospital staff, rooms, equipment). Usage of resources can cause backlogs. A DES model is first built to reflect the reality of the system it is modelling, and validation measures are conducted to ensure model accuracy. Then, adjustments to the model are implemented to understand the effect such changes would have on key performance indicators in a real-world setting. Common changes DES models consider in a healthcare setting include adding additional staff,¹⁵⁻¹⁸ beds,¹⁹⁻²¹ or equipment;^{11,15} changing scheduling policies;²¹⁻²⁴ process design;^{18,25} layout redesign;^{14,26} and the impact of changes in demand^{21,27} or discharge rates.²⁸ The application and value of using such models has been documented extensively.²⁹⁻³⁵

However, there are few studies that have applied DES models to the CT practice context, and those that do exist primarily consider a single CT machine^{9,10,13} and assess changes to patient scheduling⁹⁻¹² or appointment slot management^{13,14} processes. Previous studies that have focused on non-scheduling based changes to improve CT access times for ED patients have considered changing the proximity of CT machines to emergency rooms,¹⁴ extending staff working hours,⁵ and reducing the time to finalize the CT report.⁵ In this study, we consider three new operational interventions: hiring additional staff to transport patients between the ED and CT rooms, using a faster creatinine lab, and reducing non-value added time between when a patient's labs are completed and when transport is requested by adding a registered nurse dedicated to monitoring CT patients in the ED. Additionally, of the prior studies that consider more than one CT machine, none consider the unique scanning, resolution, and supporting equipment differences among the scanners as we do in this study.

Thus, the novelty of the current simulation study lies in the added complexity involved in modelling the combination of multiple patient

classes with the number and variety of CT machines available. Our study analyzes the effects of three interventions within the DES model that considers multiple patient classes and six unique CT machines, and draws conclusions about the efficacy of implementing said interventions on ED patient access times. Therefore, this research is performed to answer the following research question: "How can changes to the operational processes of the emergency and radiology departments improve the timeliness of service for emergency patients in need of a CT?" To the best of our knowledge, this is the first research paper to simulate the effect of accounting for multiple, unique CT machines and various patient classes when evaluating process evaluations. In this study, access times are defined as the time between when a CT order is placed and when the patient enters the CT scan room.

2 | METHODS

To test the proposed interventions before implementation, we built a DES of patient flow through the CT practice using Arena Simulation Software, Version 14.0. An overview of the patient flow through the clinical setting, data collection and analysis, and interventions is discussed next.

2.1 | Patient flow through clinical setting

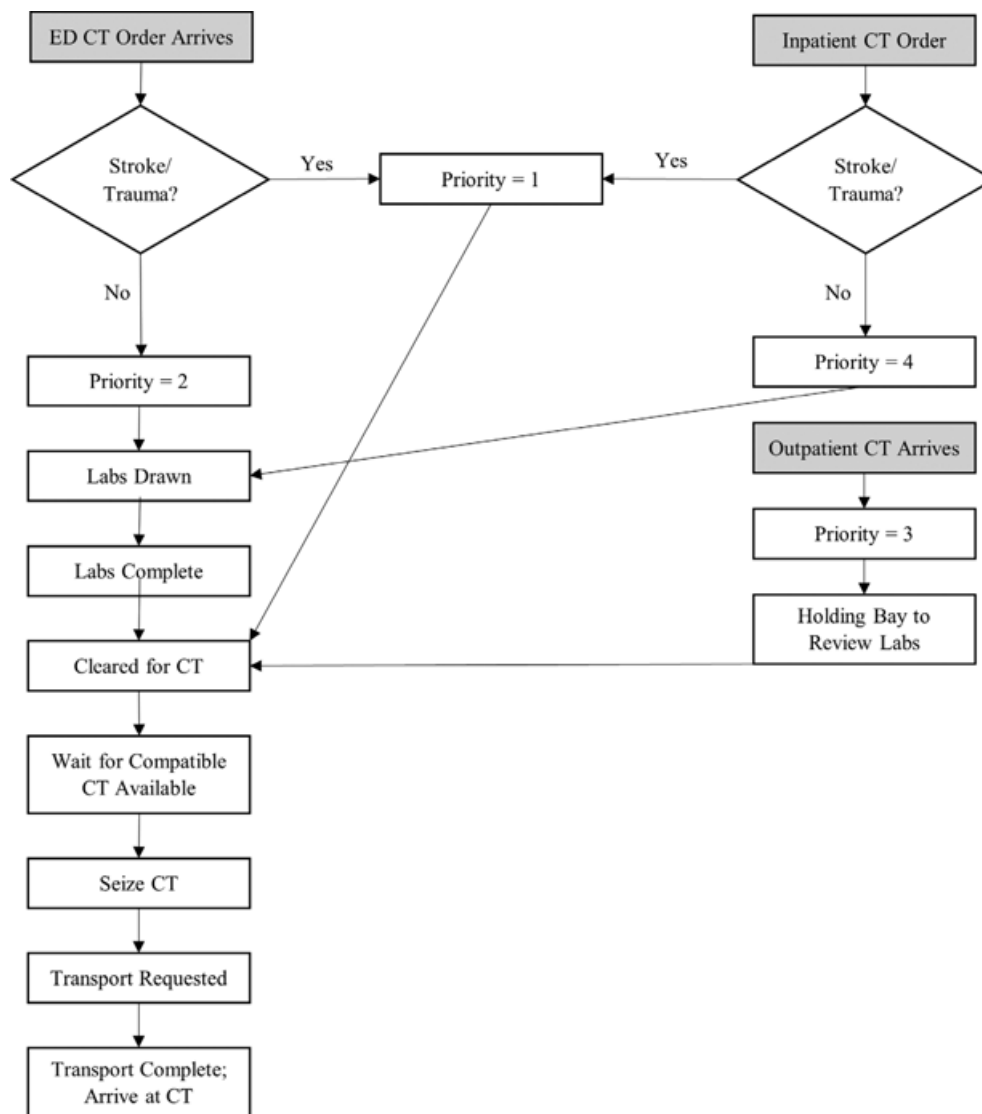
The Saint Marys Radiology department handles three types of patient arrivals: outpatients, inpatients, and emergency patients. Outpatients are scheduled to specific appointment times that vary by day of week, inpatients are semi-scheduled (ie, scans occur within a flexible time window), and emergency patients arrive unscheduled with an arrival rate that varies based on time of day. All patients have demand for the same general pool of resources: machines, holding bays (used for patient safety assessment, administering contrast and/or anaesthesia, etc.), clinic staff, and transport staff. However, these patients have numerous additional characteristics (eg, pediatric patient, presence of implanted device) and exam requirements (eg, neuro, cardiac, ablation) that dictate which CT machine(s) can safely and adequately be used, as well as how long it will take to perform the scan.

While no two individual CT machines can serve the exact same panel of patients, multiple CT machines may meet the requirements to serve any individual patient. For example, if CT1 can scan patients with characteristics A and B, and CT2 can scan patients with characteristics B and C, CT1 and CT2 are unique. Yet, a patient with characteristic B can be scanned on either CT. The process flow of each of these types of patients receiving CT scans was identified through direct observations of the CT practice and discussions with radiologists, technologists, administrators, coordinators, and other department personnel.

As such, patient flow is dependent on patient arrival type, acuity, and exam type. In this study, patients are classified according to a combination of their arrival/acuity and exam type categories as listed in Figure 1.

FIGURE 1 Patient class categorization**Patient Class =****Patient Arrival and Acuity Type:**

- 1) Trauma and/or Stroke ED
 - 2) Non-Trauma, Non-Stroke ED
 - 3) Emergent Inpatient
 - 4) Non-Emergent Inpatient
 - 5) Outpatient
- + Exam Type**

FIGURE 2 Emergency department computed tomography process flow from when an order arrives to when patient arrives at computed tomography machine

Orders for emergency patients to receive a CT originate in the ED and are categorized based on whether the patient is experiencing stroke or trauma symptoms (Figure 2). Stroke and/or trauma patients receive the highest priority and are transferred to the CT machine as soon as a transporter and compatible CT machine are available. Non-trauma, non-stroke ED CT patients are prioritized next and receive a panel of labs in the ED. After the lab results have been reviewed and a compatible CT machine is available, a request to transport the patient from the ED to the CT machine is placed. Figure 2 describes a non-trauma, non-stroke ED patient's order of events preceding a CT scan; note that trauma and/or stroke patients bypass the "labs

completed" step. Note that the prioritization used for ED CT patients is primarily trauma and non-trauma, rather than ED assigned ESI level.

Scheduled outpatients receive priority after ED patients. After checking-in at the radiology department, outpatients are transferred to a holding bay in the radiology department where a nurse reviews the patient's pre-scan labs and corresponding medical history to ensure it is safe for the patient to receive a CT. Once the patient has been approved to receive the scan and a compatible CT machine is available, a transporter is requested to bring the patient to the CT room.

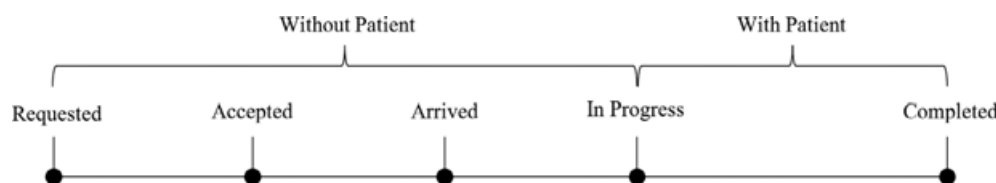


FIGURE 3 Transport process

A majority of inpatients receive a time window in which their CT exam is expected to occur and wait in their inpatient room until a suitable CT machine becomes available in this time window. The exception to this is emergency inpatient orders (eg, due to stroke) who are prioritized at the level of ED trauma and/or stroke patients. All inpatients receive pre-scan assessments in their inpatient room and are transported directly from their inpatient room to the CT machine.

All patient transfers within, into, and out of the radiology department require accompaniment by a transporter. The radiology department has access to two types of patient transporters: general and internal. General transporters serve all hospital departments. Internal transport only handles radiology department transfer requests, resulting in more predictable availability and shorter transport times. However, there are fewer internal transporters than general transporters. Transport time consists of five components (*requested*: transport request placed, *accepted*: transporter acknowledges and accepts request, *arrived*: transporter arrives at patient pick-up location, *in progress*: transporter begins transporting patient, *completed*: patient dropped off at destination) and two phases: (1) time that elapses between when the transport request was placed and when the transporter picks up the patient, and (2) active transport time with the patient (Figure 3).

2.2 | Data collection and analysis

Over 10 000 CT historical patient encounter records from April 2017 to January 2018 were obtained from EBoard, a Mayo Clinic application designed to link patient information across 15 database systems into one consolidated database and visualization system. Further information regarding EBoard and data gathering, cleaning, and consolidation is provided in an earlier manuscript.³⁶

Saint Marys CT practice offers over a thousand individual types of exams. To generate samples sizes large enough to identify distributions describing the exam length and time spent in bays, exams were grouped into 11 categories based on similar traits and the ability to be performed on the same CT machine. As seen in Table 1, exams in each category can be performed on multiple CT machines and no two CT machines are exactly the same.

For each patient category we determined the arrival rate and any restrictions on the time of day or day of week the exam could be performed (eg, outpatients do not receive CT scans at night; procedures must be performed when a physician is available). A majority of the exams are scheduled in 15-minute time slots. However, procedures and ablations take significantly longer and are scheduled for 2 and 5 hours, respectively. Additionally, we identified distributions for the

TABLE 1 CT machines able to perform each type of CT exam

Exam category	CT 1	CT 2	CT 3	CT 4	CT 5	CT 6
Abdomen/pelvis	X	X	X	X	X	X
Ablation	X					X
Biopsy	X	X	X	X		X
Cardiac				X	X	
Chest	X	X	X	X	X	X
Musculoskeletal	X	X	X	X	X	X
Neuro	X	X	X	X		X
Neurovascular				X	X	
Procedure	X	X				
Vascular	X	X		X	X	X
Trauma	X	X	X	X	X	X

time it takes to perform the exam, process the CT image after the exam is complete, and clean/prepare the CT machine for the next patient.

Analysis indicated day of week and time of day trends regarding patient arrival rates by patient classification. As such, the simulation models 24 hours, over which the outpatient and inpatient schedules vary by day of week and time of day according to historical exam schedules, and ED arrival rates fluctuate based on historical arrival data. Unique input data was used to simulate each day of the week.

Empirical transport durations were also obtained from historical data for patients transported from the ED, bays, and inpatient rooms to the CT machine rooms. Transport originating from the ED or inpatient rooms is currently served by external transporters, and transport originating in a bay is served primarily by internal transporters. The bays are located in close proximity to the CT machine rooms and, correspondingly, transport from the bays requires the shortest amount of time with a small standard of deviation (SD). On the other hand, inpatients requiring a CT can be located in inpatient rooms throughout the entire hospital, leading to the longest transportation time and SD (Table 2).

2.2.1 | Model verification and validation

A variety of verification techniques were used to ensure the conceptual description and specifications are adequately captured in the Arena model.^{37,38} The "TRACE" element and Debug features in Arena were used to verify the simulation logic, the model's logic behaviour was observed through animation, hospital administrators and ED/CT staff checked the model for accurate translation, and the effect of

perturbations to the input parameters were examined for reasonableness, including at extreme values. The baseline simulation was validated against ten months of historical electronic health record timestamp data from April 2017 through January 2018 to ensure the model was adequately representing reality. Patient (ie, entity) level validation was conducted across four metrics: (1) the number of CT orders placed, (2) the number of patients entering the CT room, (3) the number of patients exiting the CT room, and (4) the amount of time between when a CT order is placed and when the patient enters the CT room. A confidence level validation method was used to conclude the validity of the model according to the mean of each of these metrics for the three patient types using the *t* distribution and a 0.9 confidence level.³⁹ Scanner utilization was also validated for each CT machine using a *t*-test.

2.3 | Interventions

The main objective was to improve CT access time for ED patients, defined as the time from when a CT order is placed to the time the patient enters the CT room. Three process interventions were considered: adding one or more dedicated ED transporters, using an alternative creatinine lab, adding a nurse to monitor ED patient lab progress.

2.3.1 | Dedicated ED transporter

Through consultation with administrators and practitioners, six combinations of transport interventions were tested based on the number of internal transporters added (ie, 1 or 2), which shift the internal transporter would work (ie, 7a-4p, 8a-5p), and whether external transport would be requested if the dedicated internal transporter was currently unavailable (due to transporting a different patient) when a transport request arises (Table 3). If one internal transporter was hired, both policies of using or not using external transport as a backup were considered. However, if two internal transporters were hired, they would be expected to handle all ED CT transport requests without the use of external transport.

TABLE 2 Transportation time (mins)

Origin location	Observations	First quart.	Median	Mean	Third quart.	SD
Bay	903	5.0	6.0	7.1	8.0	3.0
ED	12 056	10.0	12.0	13.6	16.0	6.1
Inpatient Room	4841	20.0	25.0	27.5	33.0	11.8

TABLE 3 Transport options considered

Intervention	# of transporters	Shift	Use external as backup
Transport option 1	1	7a-4p	Yes
Transport option 2	1	7a-4p	No
Transport option 3	1	8a-5p	Yes
Transport option 4	1	8a-5p	No
Transport option 5	2	7a-4p	No
Transport option 6	2	8a-5p	No

Hypothesis: Adding internal transport staff dedicated to transporting ED patients from the ED to the CT machines will reduce transport time, thereby improving CT access time.

2.3.2 | Alternative creatinine lab

Before undergoing a CT scan, all non-trauma and non-stroke ED patients must receive a panel of labs which includes a creatinine measurement before being cleared to receive a CT scan. In 17% of these patients, the results of the creatinine lab take the longest to receive. Therefore, replacing the current creatinine lab with the faster creatinine lab test would reduce the lab wait for 17% of patients.

Hypothesis: Changing the current pre-scan creatinine lab to a faster creatinine lab test will reduce the lab time for non-trauma, non-stroke ED patients, thereby improving CT access time.

2.3.3 | Dedicated RN monitoring

Through practice observations we identified the existence of a non-value added delay after a non-trauma, non-stroke ED patient's lab results are completed; because of the multiple demands on ED nurses, it takes approximately 15 minutes from the time the labs are completed to when the ED nurse verifies the labs are completed and signals for a transport request to be made.

Hypothesis: Adding a Registered Nurse to the ED to monitor when CT ED patient labs are completed and the patient is cleared to receive a CT scan will reduce the time it takes for a transport request to be made for non-trauma, non-stroke ED patients, thereby improving CT access time.

3 | RESULTS

For each intervention, the primary performance measure is “CT Order Placed” to “CT Scan Room Enter” time, hereafter referred to as the

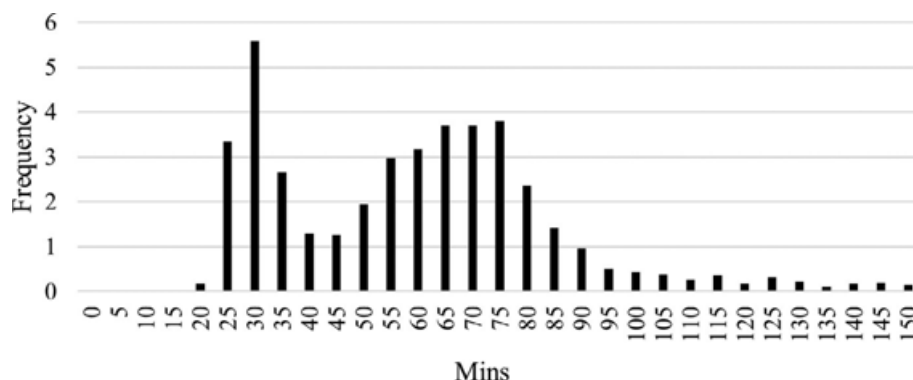


FIGURE 4 Histogram of access times (in minutes) for emergency department patients—baseline

TABLE 4 Baseline vs intervention access times for ED patients (mins)

Intervention	First quart.	Median	Mean	Third quart.	SD	Location shift [CI] ^a	Change in median	Change in mean
Baseline	33.0	58.8	61.1	73.1	34.3	-	-	-
Transport Opt. 1	24.0	46.3	47.4	61.6	25.3	-9.8 [-11.0, -8.7]	-12.5	-13.7
Transport Opt. 2	26.8	51.3	49.9	65.1	23.2	-6.9 [-8.0, -5.8]	-7.5	-11.2
Transport Opt. 3	23.0	49.4	47.9	62.9	24.8	-8.7 [-9.8, -7.6]	-9.4	-13.2
Transport Opt. 4	28.5	51.6	50.8	64.8	25.9	-6.6 [-7.8, -5.6]	-7.2	-10.3
Transport Opt. 5	22.6	48.8	45.8	61.2	21.0	-9.8 [-10.9, -8.7]	-10	-15.3
Transport Opt. 6	22.6	48.2	45.2	60.3	20.9	-10.3 [-11.5, -9.2]	-10.6	-15.9
Creatinine	31.0	53.4	57.2	69.4	32.2	-3.0 [-4.0, -2.0]	-5.4	-3.9
RN monitoring	32.5	45.1	53.3	59.0	35.1	-8.5 [-9.7, -7.3]	-13.7	-7.8

^aLocation shift parameter from Mann-Whitney test with $\alpha = 0.05$.

“access time”. Baseline access times are displayed in Figure 4 and a comparison of intervention results for ED patients are presented in Table 4. A total of 100 replications of the simulation were ran for each intervention, using a 24 hour warm-up period to ensure steady state was achieved. Metrics were captured for patients arriving within the subsequent 24 hours so that each replication corresponded to one day.

3.1 | Dedicated ED transporter

All options considered for adding dedicated internal transport for ED patients requiring a CT improved patient access time. Each option with the addition of two transporters (ie, Option 5 & 6) resulted in ED CT patients entering the CT machine on average over 15 minutes sooner than the current practice. All other options resulted in at least a 10.3 minute improvement in sample mean access time. Mann-Whitney tests with $\alpha = 0.05$ also indicate a statistically significant difference in access times for all transport options. Notably, allowing external transport to be used as a backup option if a transport request arises when the internal transporter is currently transporting another patient (ie, Options 2 and 4) performed worse than when no backup option was considered (ie, Options 1 and 3). This is likely because the time it takes for an external transporter (who could be located anywhere in the hospital) to accept the transport request and walk to the

ED (Figure 3 “Requested” to “Arrived”) is often longer than the time it takes an internal transporter to finish their current transport assignment and return to the ED.

3.2 | Alternative creatinine lab

Since trauma and stroke patients do not need to receive a panel of labs before being cleared to receive a CT scan, the results of this intervention are most relevant to non-trauma, non-stroke patients. Mann-Whitney tests indicate the creatinine intervention reduces the median access time for non-trauma, non-stroke ED CT patients who must receive their lab test results before receiving a CT (CI: [-5.02, -3.00], Table 5) as compared to the baseline access time for non-trauma, non-stroke ED patients. This also results in an improved median access time computed across all ED CT patient encounters (CI: [-4.00, -2.00], Table 4) when compared to baseline for all ED CT patients.

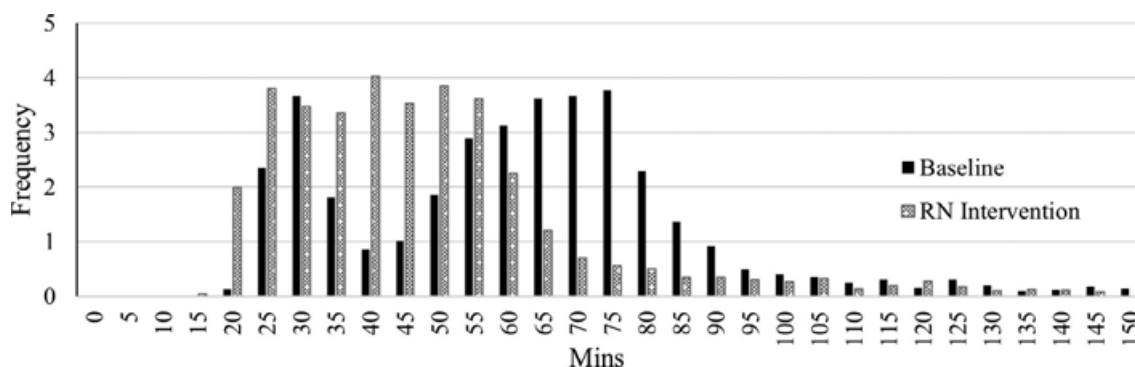
3.3 | Dedicated RN monitoring

Similar to the alternative creatinine lab intervention, the dedicated RN lab monitoring intervention primarily affects patients who require labs, that is, non-trauma, non-stroke patients. Adding a dedicated RN

TABLE 5 Baseline vs intervention access time for non-trauma, non-stroke ED patients (mins)

Intervention	First quart.	Median	Mean	Third quart.	SD	Location shift [CI] ^a	Change in median	Change in mean
Baseline	43.8	61.8	64.2	74.3	33.9	-	-	-
Creatinine	36.0	56.3	59.6	70.8	31.8	-4.0 [-5.0, -3.0]	-5.40	-4.64
RN monitoring	34.9	47.5	54.9	59.9	34.7	-11.2 [-12.3, -10.1]	-14.3	-9.3

^aLocation shift parameter from Mann-Whitney test with $\alpha = 0.05$.

**FIGURE 5** Histogram of access times (in minutes) for non-trauma, non-stroke emergency department patients

to monitor ED CT patients lab results improves median access time for non-trauma, non-stroke ED CT patients (CI: [-12.3, -10.1], Table 5) as well as for general ED CT patient access times (CI: [-9.7, -7.3], Table 4). Whereas the baseline access time distribution for non-trauma, non-stroke patients is bimodal, the RN intervention results in a unimodal distribution (Figure 5).

4 | DISCUSSION

The results indicated all three interventions improve ED CT patient access times. Overall, the interventions related to adding additional transporters resulted in the greatest improvement in access time, with an average of 15 minutes saved when two dedicated transporters are added and over 10 minutes saved with only one transporter. While the dedicated transporters can directly impact all ED patients requiring a CT, the other two interventions only directly impact non-trauma, non-stroke ED CT patients. However, they may have an indirect impact in that improving access times for any patient types may have a positive cascading effect on reducing delays for other patient types. Implementing a dedicated RN to monitor when ED patients' labs are cleared to proceed with a CT improved the median access time for non-trauma, non-stroke patients by 11.2 minutes. However, using an alternative creatinine lab only decreased median access times for this population by 4.0 minutes. Other studies have considered similar operational adjustments and found similar effects through implementation,² although no other study has used DES to preemptively analyze the effects of operational interventions in the setting of multiple, unique CT machines and various patient classes as we do in our study.

While all three of the simulated interventions suggested improvements in ED patient access time, after conducting an internal cost-benefit analysis of the interventions, the practice initially implemented dedicated ED transporter option 1: adding one transporter from 7a to 4p with external transport backup. Furthermore, the radiology operations manager stated that "this was an easy transition fully supported by Radiology leadership due to low cost versus impact." Results have materialized in practice. In addition to meeting the 25 minute door-to-imaging guidelines for stroke patients presenting to the ED, Mayo Clinic also has an internal goal of having all ED CT patients on the CT table within 30 minutes from the time the CT is ordered. The initial implementation results of this simulation study indicate that the practice has seen a 10% reduction in access delays and now hit their 30 minute access goal at least 80% of the time.

One of the major limitations of this study is that the ED, CT, and transporter staff were aware of the Radiology Clinical Learning Laboratory, this study, and its intent to improve access times. The ongoing interactions with care staff and tracking of metrics throughout this study may have affected their behaviour and contributed to some of the benefits seen. Additionally, as with any simulation, it is not possible to capture all of the details and variance in processes that happen in a large academic medical center. It was our intent to capture as many of the nuances as possible while still retaining model tractability. Finally, other institutions may have different process flows or define access time differently than we considered in this study, which may reduce the generalizability of this work.

Simulation provides stakeholders the opportunity to quickly evaluate many different process interventions with no detriment to the practice or patient experience. In this study, we built a DES model that is accurate to the current state of patient flow within the CT practice at large academic medical center using distributions based on

empirical data. Once validated, the simulation was manipulated by imposing strategic process interventions hypothesized to improve patient access times, and simulation results illustrated the effect of such interventions on CT operations.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Kayse Lee Maass, Mustafa Y. Sir, Michelle R. Nordland, and Kalyan S. Pasupathy were responsible for the study design. The simulation model was primarily created by Kayse Lee Maass with assistance from Elizabeth Halter and Todd R. Huschka. Kayse Lee Maass and Elizabeth Halter were responsible for the data analysis. Kayse Lee Maass, Todd R. Huschka, Mustafa Y. Sir, Michelle R. Nordland, and Kalyan S. Pasupathy contributed to interpreting results and contributions to practice. Manuscript preparation was primarily completed by Kayse Lee Maass. All authors reviewed the paper prior to submission.

ETHICS STATEMENT

No ethical approval required for this study.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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